Computer Science & IT

Theory of Computation



TM

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

TM

4 hours

$a^n b^m c^k d^l$ $n+k = m+l$ → CFL

1hr →

TM Can accept → (Reg, CFL, CSL, Rec, RE)

Resources?

2 hour + 2 hour + 2 hour

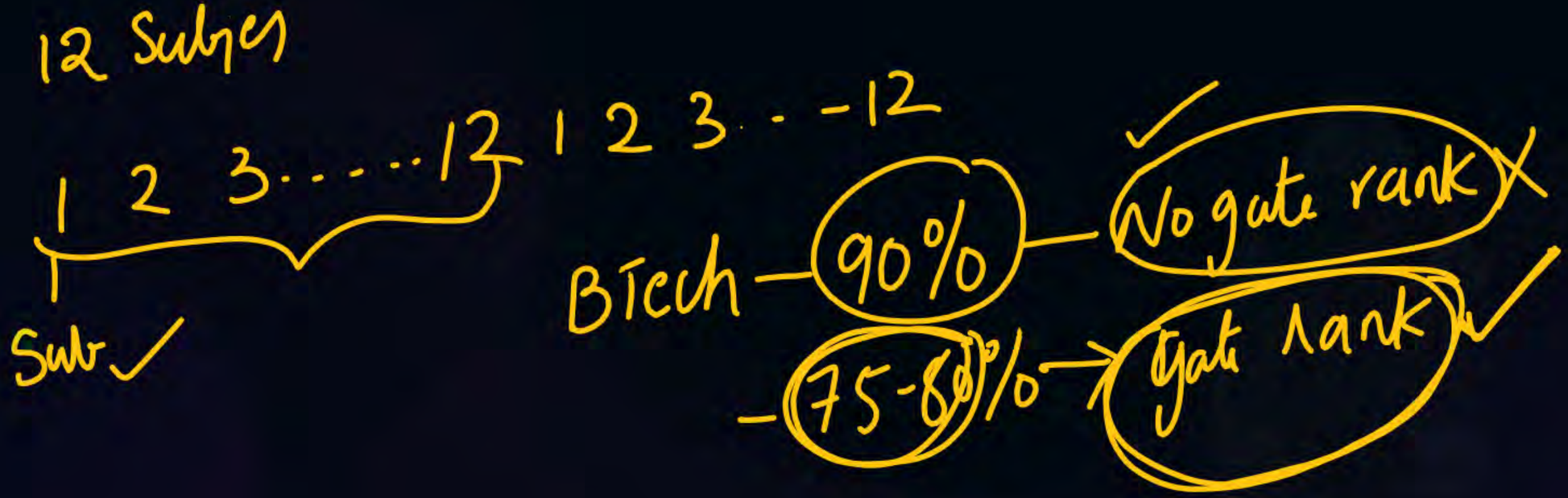
maintain TOC? (How to make short notes)

Revise

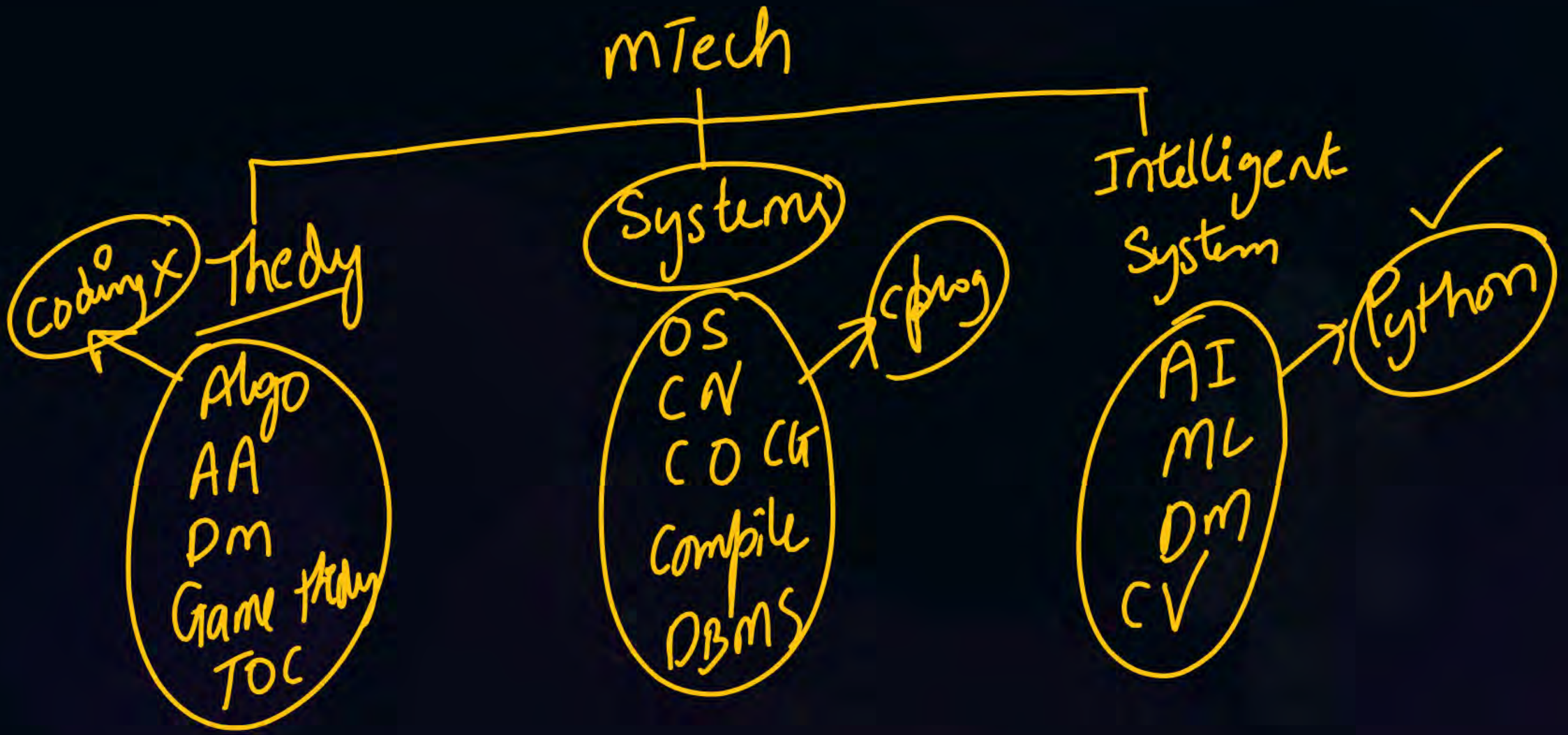
AIR < 100

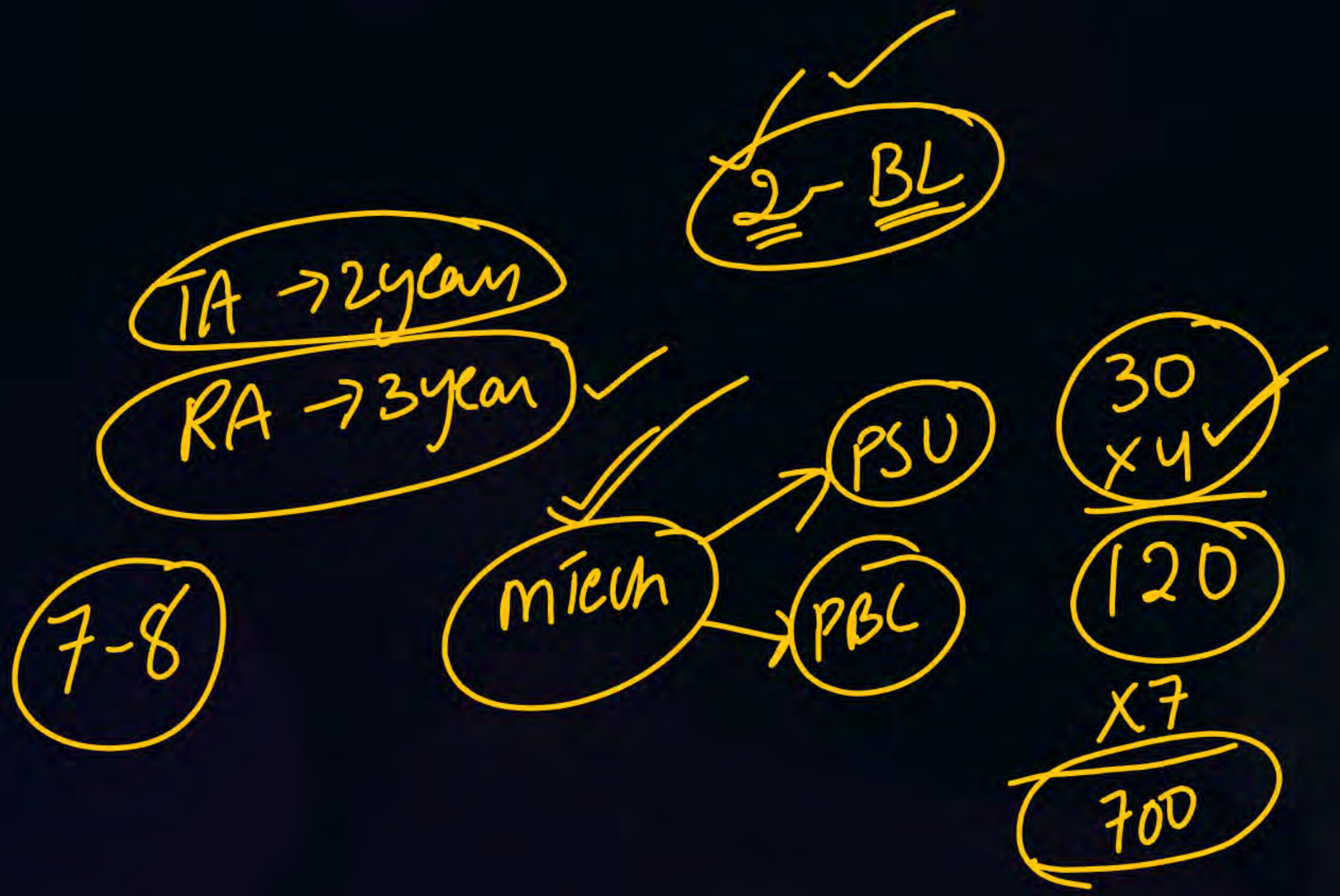
DFA, NFA

RE


12 Subjects
1 2 3 12
1 2 3 . . 12
Sub
BTech
90%
No gate rank
75-80%
Gate rank

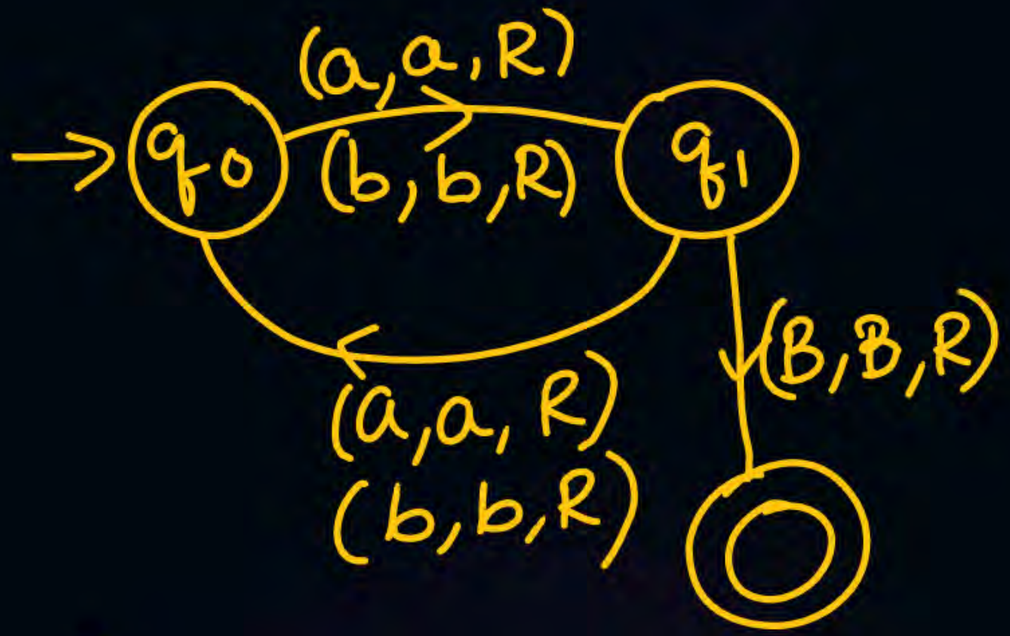

1) Gate
2) Problem Solving
PBC
CP
Java
C++
Python
Algotos
leet code
101
2
Gate
AI + ML
Data Scientist
ML Eng
Python


mTech
Systems
Intelligent System
Theory
Coding
Algo
AA
DM
Game Theory
TOC
OS
CN
CO CG
Compile
DBMS
cprog
AI
ML
DM
CV
Python


2 yr BL
TA → 2 year
RA → 3 year
7-8
M Tech
PSU
PBC
30
x 4
120
x 7
700

Formulae :-





$\epsilon \rightarrow$ halt, Rej

ϵ

a

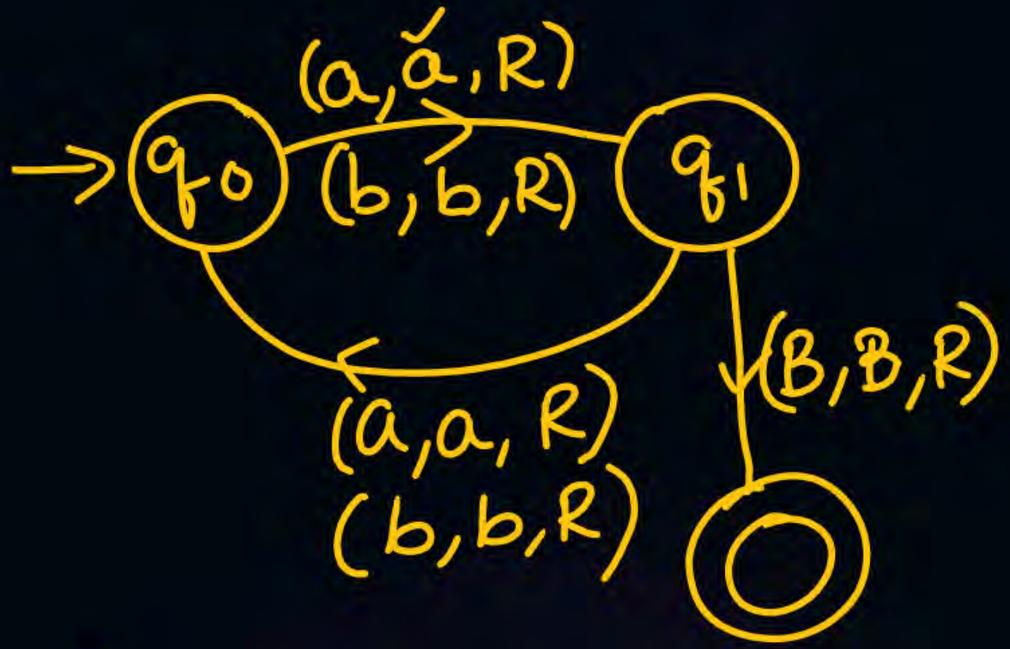
b

aa

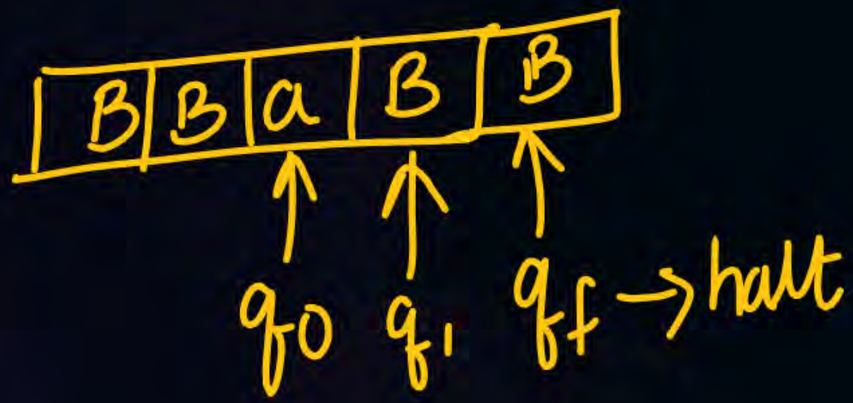
bb

aaa



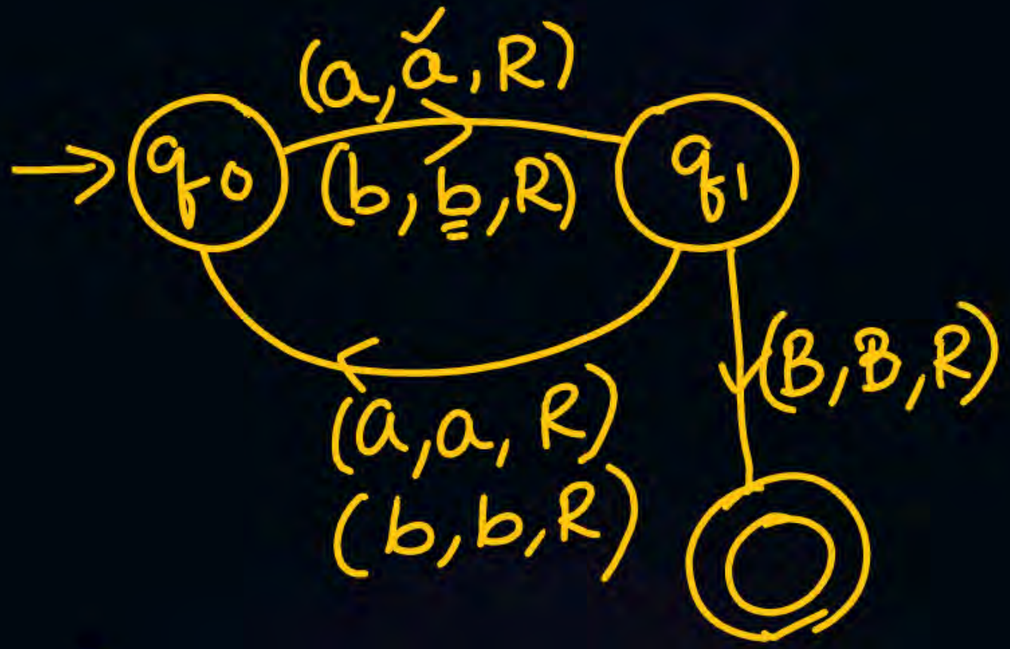


ϵ
a
b
aa
bb
aaa



$\epsilon \rightarrow$ halt, rej

$a \rightarrow$ halt, acc



ϵ

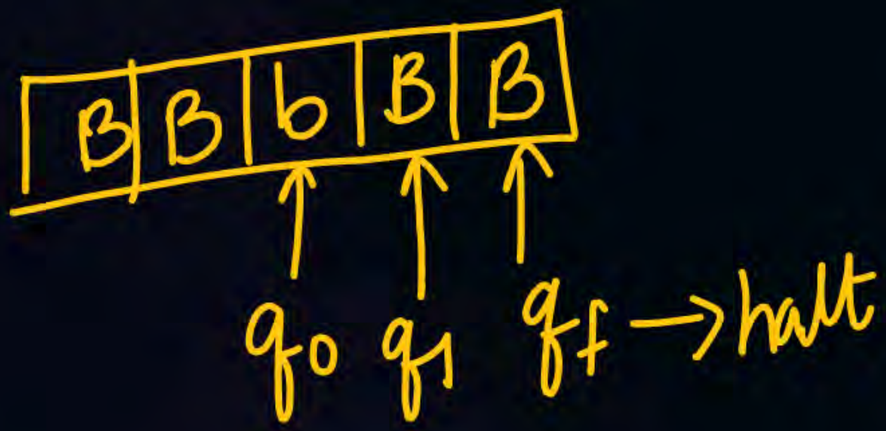
a

b

aa

bb

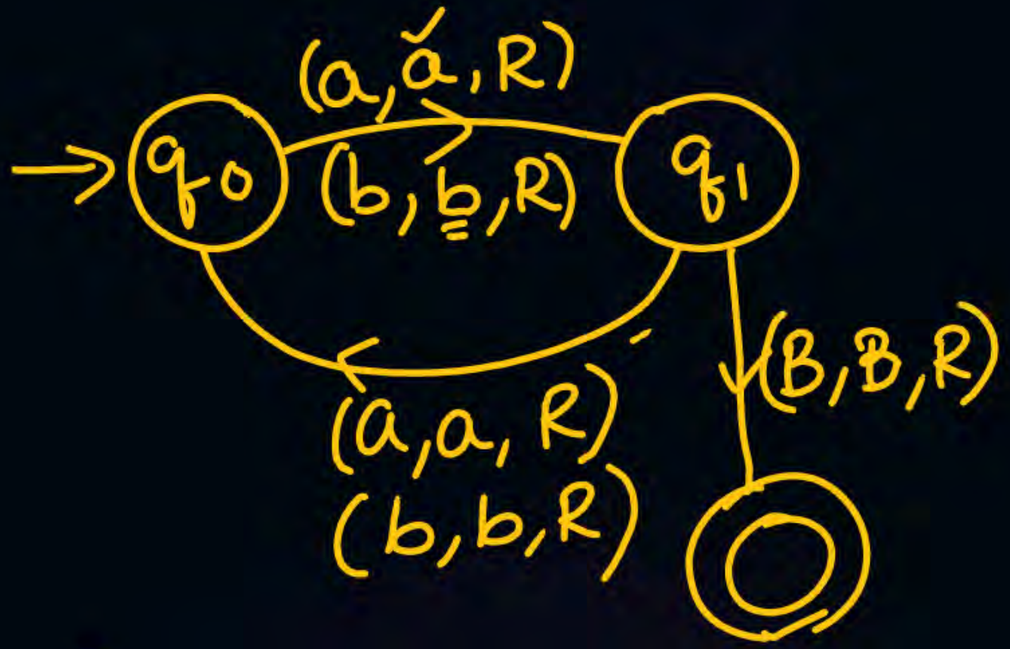
aaa



$\epsilon \rightarrow$ halt, rej

a → halt, acc

b → halt, acc



ϵ

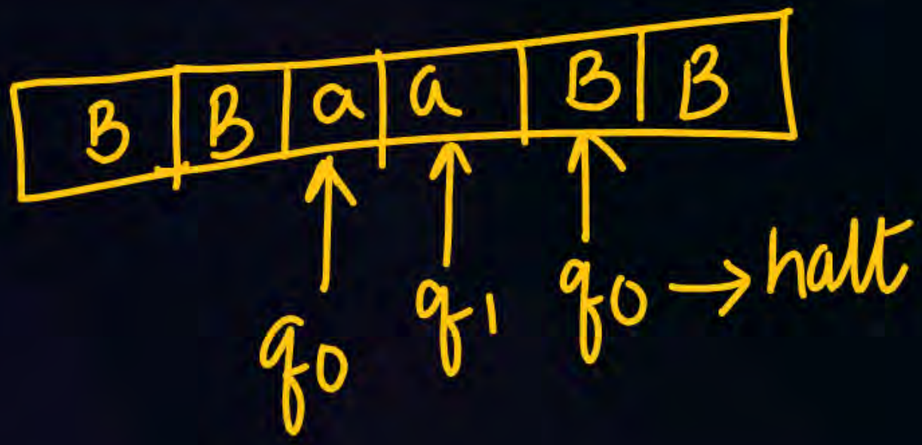
a

b

aa

bb

aaa

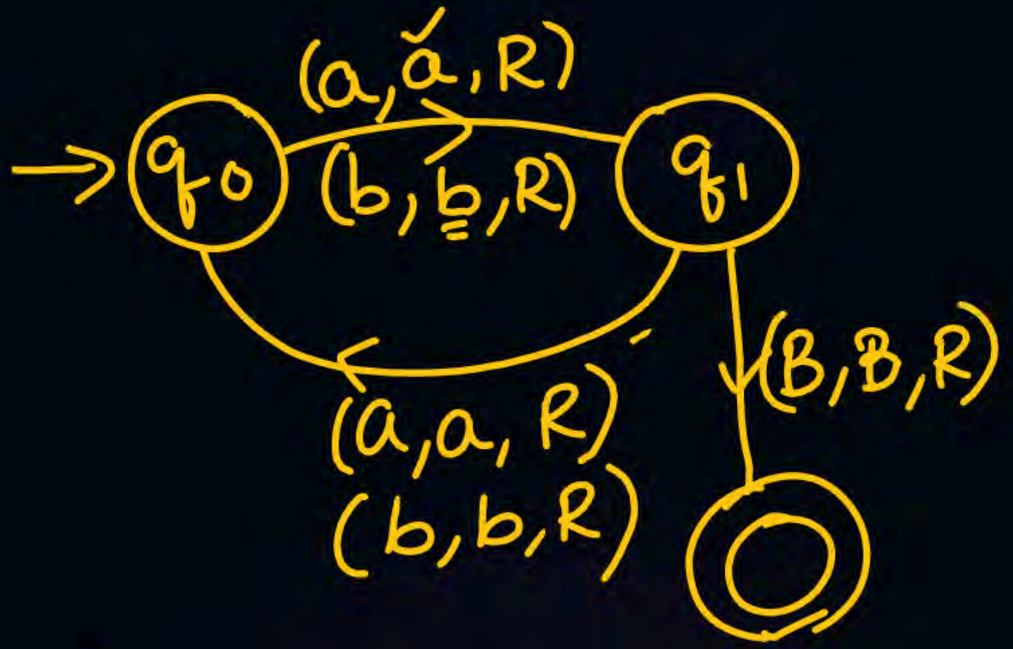


$\epsilon \rightarrow$ halt, Rej

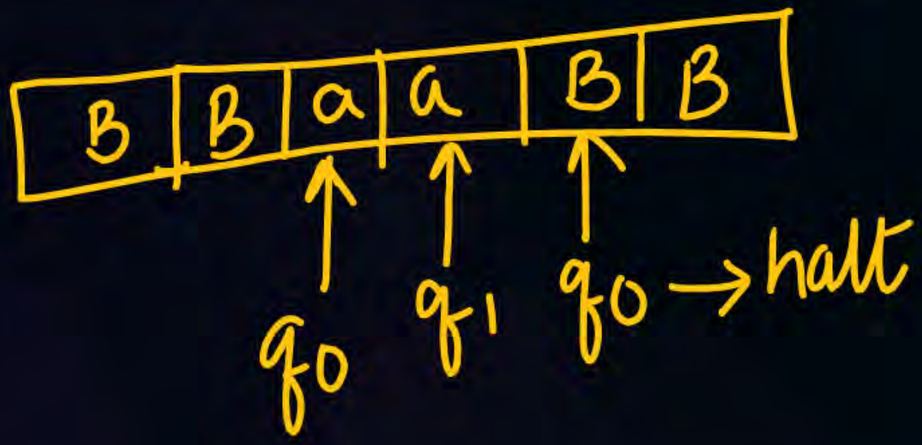
$a \rightarrow$ halt, acc

$b \rightarrow$ halt, acc

$aa \rightarrow$ halt, Rej



ϵ
a
b
aa
bb
aaa

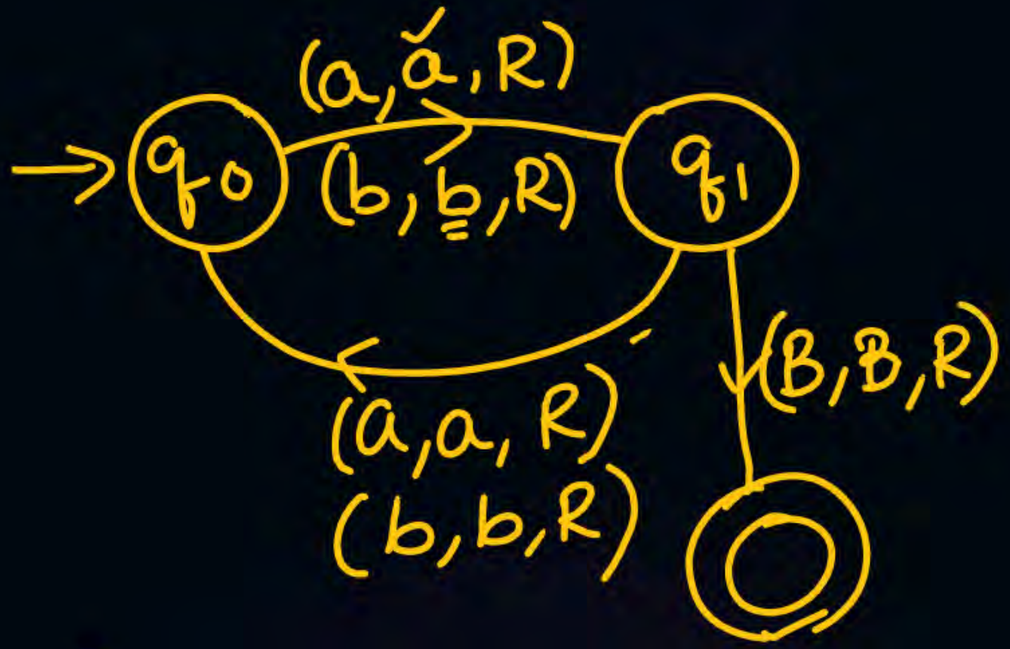


$\epsilon \rightarrow$ halt, Rej

$a \rightarrow$ halt, acc

$b \rightarrow$ halt, acc

$aa \rightarrow$ halt, Rej



ϵ

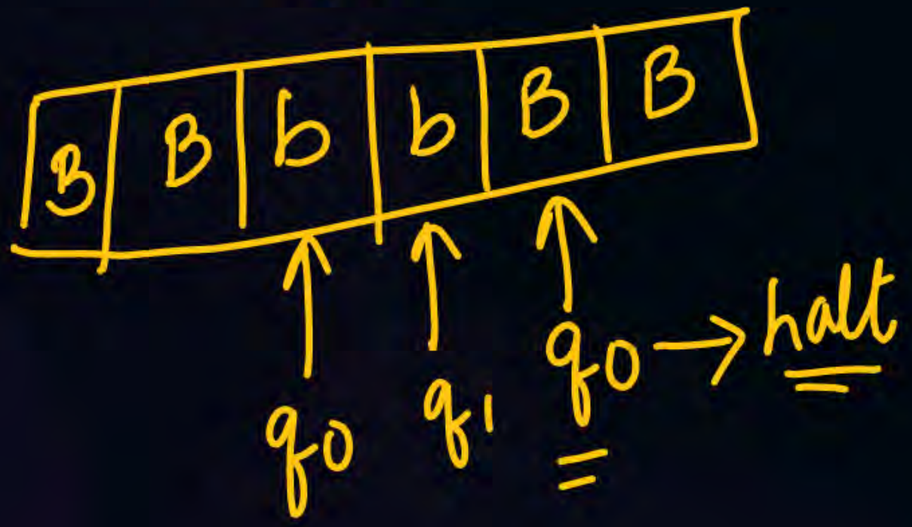
a

b

aa

bb

aaa



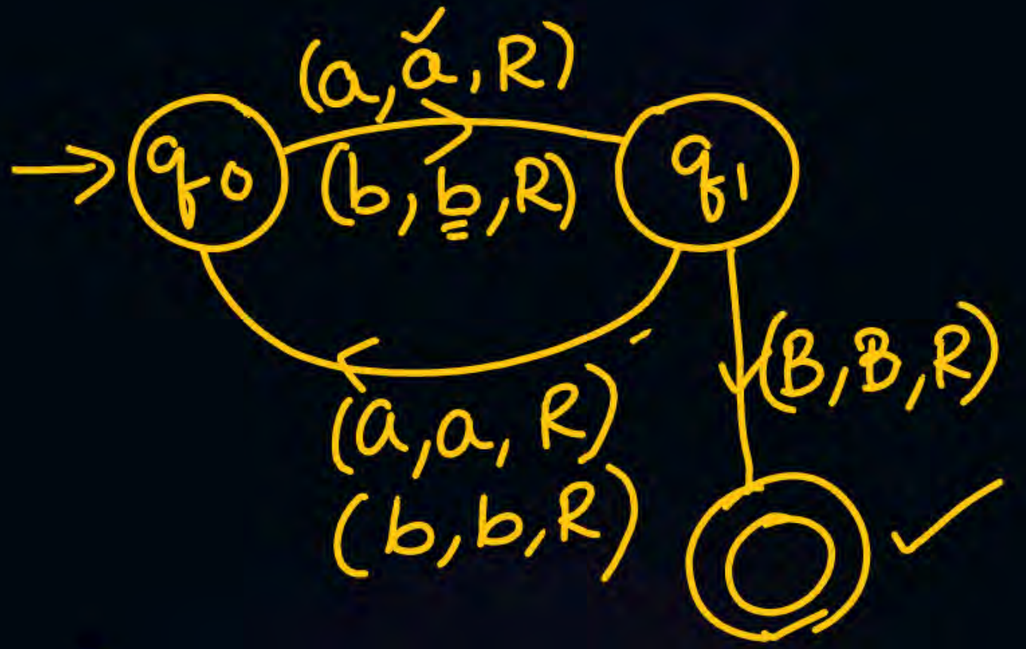
$\epsilon \rightarrow$ halt, Rej

$a \rightarrow$ halt, acc

$b \rightarrow$ halt, acc

$aa \rightarrow$ halt, Rej

$bb \rightarrow$ halt, Rej



$L = ((a+b)(a+b))^*(a+b)$

halts $\rightarrow (a+b)^*$

hang $\rightarrow \phi$

ϵ
a
b
aa
bb
aaa

B	B	a	a	a	B	B

$q_0\ q_1\ q_0\ q_1\ q_f \rightarrow$ halt

$\epsilon \rightarrow$ halt, Rej

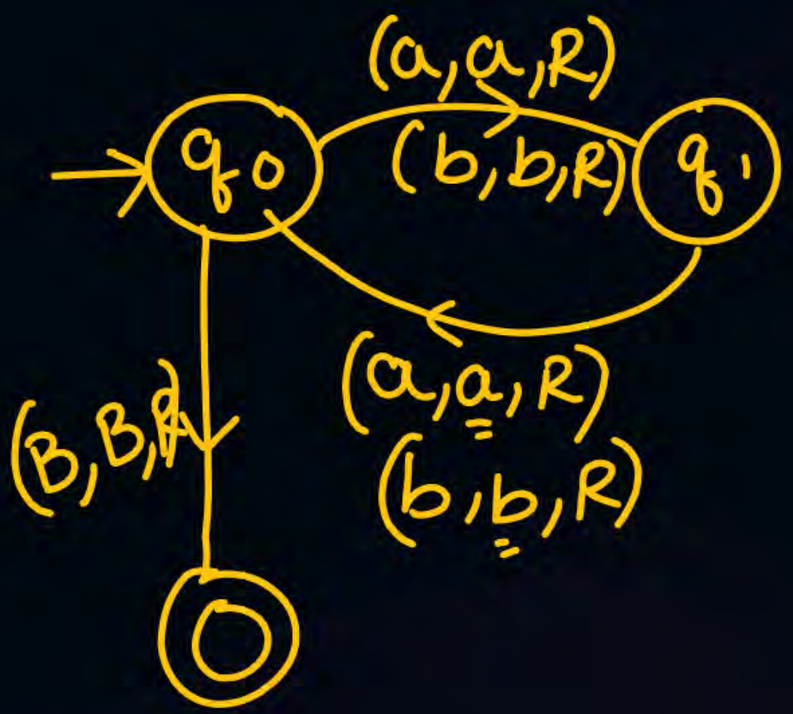
a → halt, acc

b → halt, acc

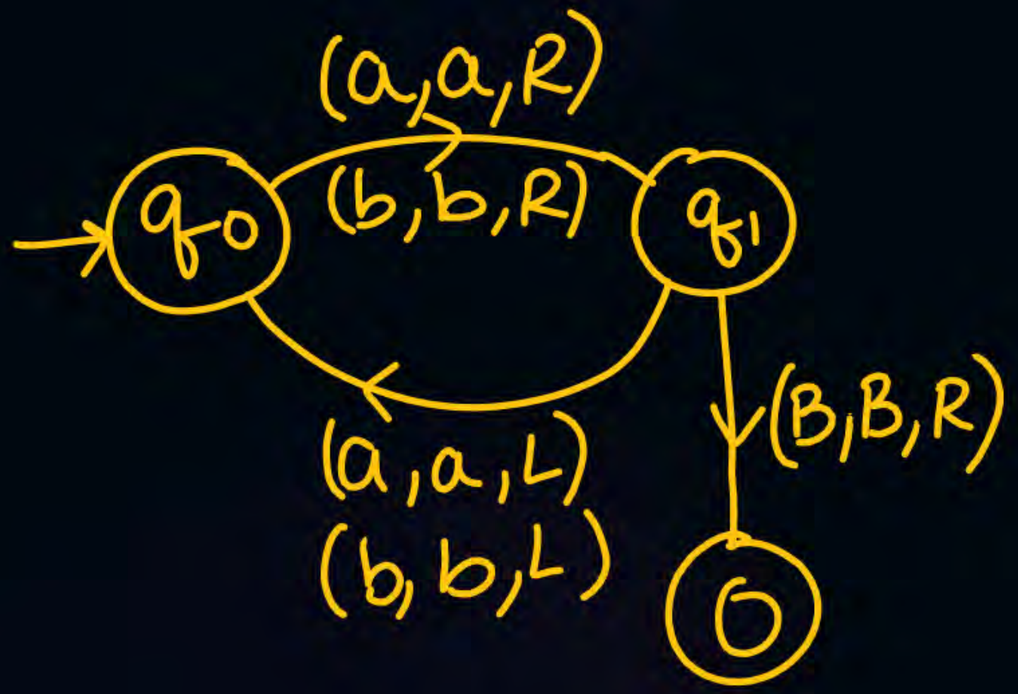
aa → halt, Rej

bb → halt, Rej

aaa → halt, acc



ϵ
a
b
a b }
a a


(a,a,R)
(b,b,R)
q_0
q_1
(a,a,L)
(b,b,L)
(B,B,R)
O



2 mins Summary

Topic Turing Machines

Topic

Topic

Topic

Topic

THANK - YOU